

Direction, not magnitude

a differentiable surrogate's largest design gradient can be confidently sign-wrong

— a solver-free gate catches it by cross-seed sign agreement, not magnitude

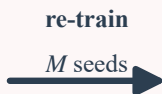
THE BIGGEST GRADIENT

large gradient



re-train

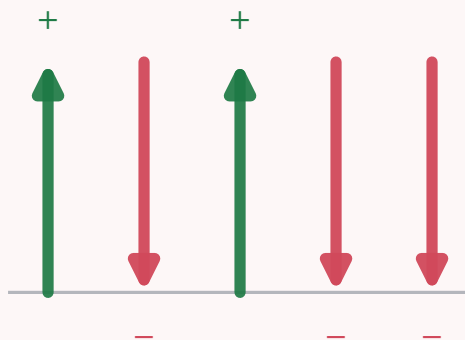
M seeds



same design

can be confidently *WRONG*

signs **FLIP** across seeds



cross-seed sign disagrees



magnitude rule
TRUSTS it



our gate
REJECTS it

A SMALLER, SIGN-STABLE GRADIENT

the impedance gradient: tiny, but every seed agrees



signs **AGREE** across seeds



TRUSTED

use free autodiff

trusts impedance, rejects the seed-unstable radiation gradient

direction (sign), not magnitude, decides trust

AUC 0.91

gradient sign-correctness
external transfer-matrix benchmark



Solver-free gate

no full-wave solve to decide trust



Signs agree

free autodiff gradient (0 solves)



Signs disagree

one full-wave solve, only here